Q-volution Hackathon 2026

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Sponsor: Quandela (Tier 2 partner)
Event Date: **February 23rd to March 1st, 2026**

Instructions:
Please fill out the sections below to define your challenge track. This information will be used to populate the hackathon website and guide participants during the "Sprint" phase.

Challenge Details

Challenge Title:
Option pricing in finance

One-Sentence Tagline:

Through this prompt, you will explore how quantum circuits can be used to **price options** a **quantum machine learning** (QML) model.

Problem Statement / Description:
(Please provide 1-2 paragraphs describing the scenario. What is the real-world problem? What specific task are the students trying to solve?)

Quantum Machine Learning Problem. Quantum Machine Learning provides an alternate method by learning the distribution of future option prices from historical data. For example, quantum reservoir computing (QRC) is a type of hybrid model. In this model, a fixed quantum circuit extracts non-linear features from the data and feeds them to a trainable classical algorithm.

Your task is to use QML to produce an option pricing model. More specifically, your goal is to implement and train a QML model to predict the price of put and call options. A training dataset will be provided to the participants to train a QML model to price options. The test dataset will be provided at the end of the hackathon for fair and exciting evaluation.

Difficulty Level:

[x] Beginner (Focus on education/implementation) – **a ML background is preferred**

[x] Intermediate (Focus on optimization/creativity)

[x] Advanced (Focus on novel approaches/research)

We believe participants of all backgrounds can build small models using MerLin. The platform provides multiple API entry points: some geared toward ML practitioners, others toward those more comfortable with quantum concepts. It should allow everyone to engage at their own level of expertise.

Technical Requirements

Required Tools/SDKs: MerLin by Quandela (pip install merlinquantum)

Specific Technical Constraints (optional):
In simulation: up to 20 modes and 10 photons

On Quandela's QPU:

- up to 24 modes and 12 photons
- does not support amplitude encoding and state injection

Learning Resources (for the bootcamp)

We have a "Bootcamp" phase (Mon-Thu) before the hacking begins. Please list resources students should study to prepare for your challenge.

Tutorials / Sample Code:

Ressource	Description
Colab notebook on Pytorch	Initiation notebook to **Machine Learning** concepts and **PyTorch** basic usage
Training center : Quantum computing hackathon	Step by step lessons adapted to ML and quantum people and targeting the **knowledge needed to tackle this challenge** !
A ready to run MerLin tutorial	A tutorial presenting **MerLin**, step by step
Quantum Reservoir Computing for Realized Volatility Forecasting	An **example of QML model** on a similar subject
MerLin documentation	Documentation on the **MerLin** framework
A paper from our first hackathon	A paper presenting the **approaches that made it to the final phases** of our first Perceval Quest hackathon A GitHub repository with all code is available

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Judging Criteria

How should we evaluate the projects? Please assign a rough weight to these categories or add your own.

- **Accuracy/Technical Merit:** 35 %
- **Creativity/Novelty:** 35 %
- **Presentation/Documentation:** 30 %

Workshop Session

You have a 45-minute technical workshop slot. Who will be leading this session?

Speaker Name: **Cassandre Notton**
Topic/Title: **Quantum Machine Learning with Merlin**
Preferred Date/Time in ET (during bootcamp Mon-Thu):
Mon/Tuesday/Wednesday online and in the afternoon in Europe (the speaker being in Montreal)